Name: Date: Period:

Power

Notes

Power:

Units:

Horsepower:

Practice

1. A cable lifts a 2600-kg elevator a distance of 30 m in a time of 8 seconds.
   (a) What power does the motor output in accomplishing this task? Express in horsepower.

   (b) If the time to lift the elevator doubled to 16 seconds, what would the power output be?

   (c) In which case was more work done: question (a) or question (b)? Explain.

2. Two cars travel 12 km. They travel against identical friction and air drag forces totalling 2500 N. Car A travels at a constant speed of 30 m/s; Car B travels at a constant speed of 45 m/s.
   (a) Which car's engine exerted more work?

   (b) Which car's engine had a greater output of power?

   (c) Calculate each car's power output in horsepower.

3. A compact fluorescent 60-watt replacement bulb only requires 13 W of power. Over the course of an hour, how many Joules of energy are saved?

4. You get an electric bill for 50 kWh (killowatt-hours). How much electrical work was done by your house in Joules.